

The Measured Anatomy of AI’s Economic Impact: Task-Level Evidence from Usage Data

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Abstract

Every input of the standard calculation of AI’s aggregate productivity potential can now be measured rather than imputed. Using task-level usage records covering 2,063 O*NET tasks across five data vintages, we measure time savings per AI-assisted task instance (90% at the mean), task success rates (76%), the share of usage that is market work (50% on a wage-bill basis), and the share of usage delegated rather than supervised (43%). An accounting framework maps these into aggregate TFP: at current adoption incidence, the realized gain is bounded between 0.6% and 1.4%, and [Acemoglu \(2025\)](#)’s well-known 0.66% upper bound emerges as the conservative corner of the measured calculation rather than a competing estimate. Three further results discipline how the gains will arrive. Cost cannot be the binding constraint: median human-to-AI cost ratios exceed 600, and a price collapse of 92% in commodity-tier model prices moves task-level usage composition by approximately nothing. Deployment infrastructure is the near-term frontier: within the same task, scaffolded API usage is 30 percentage points more automated than interactive consumer usage, a measured headroom largest precisely where outputs are *not* machine-checkable. The binding constraints on AI’s productivity contribution are, in order: adoption incidence, deployment infrastructure, and verification — not model capability, and not price.

JEL: O33, O47, J24, E24

Keywords: artificial intelligence, total factor productivity, task-based framework, technology adoption

1 Introduction

How much will artificial intelligence raise aggregate productivity, and what currently limits it? The most influential answer comes from [Acemoglu \(2025\)](#): a task-based calculation multiplying the share of work exposed to AI ([Eloundou et al., 2024](#)), the share of exposed tasks profitably performable (a computer-vision-specific estimate; [Svanberg et al., 2024](#)), and an average cost saving per task taken from two early experiments ([Noy and Zhang, 2023](#); [Brynjolfsson et al., 2025](#)). The result — an upper bound of 0.66% TFP over ten years — rests on inputs that were, at the time, necessarily imputed: exposure scores predicted from task descriptions, and cost savings extrapolated from two settings.

Every one of those inputs can now be measured. Anthropic’s Economic Index ([Handa et al., 2025](#); [Anthropic, 2026](#)) classifies millions of AI conversations into O*NET tasks and reports, per task: usage intensity, estimated completion time with and without AI, task success, the purpose of use (market work, personal, or coursework), and the mode of interaction (delegation or collaboration). We assemble five public vintages of these data (December 2024 through February 2026), merge them to O*NET task characteristics and occupational wages and employment, and rebuild the aggregate calculation from measured micro-inputs — together with an accounting framework that makes explicit which corrections compose and which are alternatives.

Three numbers organize the paper. First, measured time savings per AI-assisted task instance are enormous: 90% at the mean (a 9–12 $\times$ speedup), an order of magnitude above the 27% average that [Acemoglu \(2025\)](#) took from the experimental literature. Second, the aggregate implication is nevertheless small: passing the measured savings through measured success rates, the measured market-work share, the measured delegation share, and current adoption incidence yields a realized TFP gain between **0.6% and 1.4%**. The distance between these two numbers is not a contradiction; it is an anatomy. Each wedge between them is measured, and the largest ones are not the ones the literature emphasizes: more than a third of the wage-bill-weighted usage is not market work at all, and roughly half of work usage requires in-loop human collaboration. Third, [Acemoglu \(2025\)](#)’s 0.66% sits inside our band — indeed, replacing our measured per-instance savings with the most conservative field-experimental estimate ([Brynjolfsson et al., 2025](#)) reproduces his number almost exactly. The measured calculation does not overturn the back-of-the-envelope one; it explains, term by term, why the envelope was about right, and shows which terms will move.

We are explicit about what these data can and cannot identify. Usage records condition on self-selected adoption: users bring AI the tasks — and the parts of tasks — where it works. Anthropic’s own documentation is frank about this (“users bring tasks to Claude that they’re more confident will work”). We therefore bound rather than estimate. The full-usage ceiling implied by per-instance savings (39% of the wage bill, absurd on its face) is reported not as a forecast but as a diagnostic: its distance from any credible aggregate is a measurement of selection. The lower bounds substitute average treatment effects from randomized and field experiments for our per-instance savings; because those experiments measure realized output in actual work settings, they already embed the quality and verification wedges, and the framework treats them as an alternative

correction rather than stacking corrections twice.

The remainder of the paper disciplines *how* the gains will arrive, with three results on the binding constraint.

Cost no longer binds. The median task in our data costs over 600 times more to perform with human labor than with AI inference. In the cross-section, tasks with larger cost advantages are used more — but the gradient is absorbed entirely by a single capability control, because cost advantage and capability are collinear across tasks. The panel settles it: between vintages, commodity-tier model prices collapsed by 92% while frontier prices were flat, generating differential price paths across tasks; in first differences — where fixed task capability nets out — the price elasticity of usage composition is -0.009 , with a design powerful enough to detect 0.031. The historical elasticity of technology diffusion with respect to adoption costs is approximately -1 (Comin and Hobijn, 2010; Anonymous, 2026); AI usage today is more than thirty times less price-sensitive than that benchmark. AI is the first technology in this literature to launch *past* its cost margin, and the cost-diffusion law correctly goes quiet there.

Deployment infrastructure is the near-term frontier. Comparing the same O*NET task across the two platforms in the same week, scaffolded API usage is 30 percentage points more automated (delegated rather than supervised) than interactive consumer usage — 85% versus 47% in aggregate, with 89% of common tasks more automated on the API side. We frame this gap as measured *headroom*, not a treatment effect: production deployments select their tasks and instances. But its anatomy is informative. A shift-share decomposition shows the consumer delegation share is not falling because of task composition; it falls *within* tasks, while on the API the within-task complexity gradient runs the opposite direction — bigger jobs become *more* delegated where scaffolding absorbs them.

Verification is bought wholesale, not task by task. On the consumer platform, delegation follows machine-checkability exactly as theory predicts: 58% automation for verifiable tasks (code, queries, calculations), 43% in the middle, 32% for unverifiable tasks (advising, teaching, negotiation). But the API-consumer headroom is *largest* for unverifiable tasks (+44 points versus +26), and the API gradient is nearly flat. Deployment context substitutes for machine-verifiability: production harnesses delegate broadly, with verification moved downstream or into process, rather than expanding delegation only where outputs can be checked mechanically. Machine-verifiable tasks are only 15% of the wage bill; the productivity action is in the unverifiable middle mass — which is exactly where the deployment headroom operates.

Related literature. The paper speaks to four strands. First, aggregate AI projections: Acemoglu (2025) provides the framework we populate with measured inputs; our band brackets his bound and identifies which input each side of the band moves with. Second, the experimental literature on AI productivity (Noy and Zhang, 2023; Brynjolfsson et al., 2025; Peng et al., 2023; Dell’Acqua et al., 2023): we use these estimates as the conservative corner of a partial-identification interval, and we reconcile the order-of-magnitude gap between experimental ATEs (15–56%) and

usage-measured per-instance savings (90%) as selection into usage — uniformly across the very domains the experiments studied. Third, measured adoption: [Bick et al. \(2024\)](#) (1–5% of work hours AI-assisted; time savings 1.4% of hours), [Bonney et al. \(2024\)](#) (firm adoption), and [Humm and Vestergaard \(2025\)](#) (3% time savings among Danish users; no detectable earnings effects). These independent measurements cohere with ours once the incidence margin is explicit: 90% per-instance savings on 1–5% of hours is 0.9–4.5% of work time, bracketing both survey estimates; and our realized band of 0.6–1.4% maps directly to [Bick et al.](#)’s 1.4%-of-hours figure at the labor share. Fourth, usage-record measurement itself ([Handa et al., 2025](#); [Chatterji et al., 2025](#)): [Chatterji et al.](#)’s finding that ChatGPT’s non-work share has grown beyond 70% independently corroborates the non-market wedge we measure for Claude (50% of wage-bill-weighted usage), which to our knowledge has not previously been propagated into an aggregate productivity calculation.

Section 2 sets out the accounting framework. Section 3 describes the data, merges, and validation. Section 4 shows cost no longer binds. Section 5 presents the waterfall and the bounds. Section 6 measures the deployment headroom. Section 7 presents the verifiability gradient. Section 8 discusses external validity. Section 9 concludes.

2 Accounting Framework

The framework is [Acemoglu \(2025\)](#)’s application of [Hulten \(1978\)](#), extended to make usage-based measurement explicit. Tasks j enter production with wage-bill shares θ_j (so $\sum_j \theta_j$ equals labor’s share of costs in covered work). When AI performs a share of task j ’s work, the aggregate TFP gain is

$$\Delta\text{TFP} = s_L \sum_j \tilde{\theta}_j \iota_j \sigma_j, \quad (1)$$

where $\tilde{\theta}_j$ is task j ’s share of the total wage bill, $\iota_j \in [0, 1]$ is the *incidence* of AI assistance in task j ’s work, σ_j is the *effective* cost saving per assisted unit of work, and s_L is the labor share converting labor-cost savings into total-cost savings.¹

The content of the paper is the measurement of σ_j and ι_j . Usage records provide a per-instance saving s_j : one minus the ratio of estimated completion time with AI to estimated time without, on the instances users actually brought to the model. Three measured wedges separate s_j from the effective saving σ_j on assisted work:

$$\sigma_j = s_j \times q_j \times \omega_j \times \delta_j, \quad (2)$$

with q_j the task success rate (failed attempts save nothing), ω_j the share of task j ’s usage that is market work (personal and educational usage generates surplus but not measured output), and $\delta_j \in [a_j, 1]$ the verification correction, bounded below by the share of usage in delegation modes a_j (in collaborative modes the human remains in the loop for the duration, and we cannot observe how

¹[Acemoglu \(2025\)](#) implements the same conversion with industry labor shares (his 27% labor saving becomes 15.4% total). We use $s_L = 0.6$ throughout.

much of the measured saving survives; we therefore report a band, crediting collaborative-mode savings fully at the upper bound and not at all at the lower). Two estimators of ΔTFP follow:

Path A (usage-measured). Insert measured $s_j q_j \omega_j [a_j, 1]$ into (2) and a common incidence ι into (1). This is the waterfall of Section 5; its output is a band, from the verification bound.

Path B (experiment-anchored). Replace $s_j q_j \delta_j$ wholesale with an experimental average treatment effect $\bar{\sigma}^{\text{ATE}}$. Field and lab experiments measure realized output in actual work settings: failed attempts, rework, and in-loop collaboration are already inside the treatment effect, and the work context disposes of ω_j . Path B therefore substitutes for the quality and verification wedges rather than composing with them — stacking both would double-count.²

Selection into usage is the identifying caveat of Path A and the reason Path B exists. Every s_j is an adopter-and-instance-conditional quantity. We treat the resulting ceiling (set $\iota = 1$, $\delta = 1$) as a diagnostic of selection, the band at measured incidence as the headline, and Path B as the conservative anchor. Incidence ι is the single most consequential free parameter; rather than committing to one construction we report $\Delta\text{TFP}(\iota)$ as a function with externally measured candidates marked (Bick et al., 2024; Bonney et al., 2024).

3 Data and Measurement

3.1 Sources and construction

Usage records. The Anthropic Economic Index (AEI) publishes privacy-preserving classifications of Claude conversations (Handa et al., 2025; Anthropic, 2026). We use five vintages: two early task-share releases (data windows around December 2024–January 2025 and February–March 2025) and three full-facet releases (single weeks in August 2025, November 2025, and February 2026), each covering both the consumer product (Claude.ai) and first-party API traffic. The November 2025 consumer week is our cross-sectional baseline: 3,170 O*NET tasks with usage shares, estimated human-only and with-AI completion times, success classifications, use-case splits (work/personal/coursework), and interaction-mode splits (delegation modes: directive, feedback loop; collaboration modes: task iteration, learning, validation).

Task characteristics and wages. AEI task names map to O*NET task statements by exact normalized text (86.3% merge rate; unmatched names are predominantly O*NET version drift — median fuzzy-match similarity 0.96 to the 30.1 statements — not new tasks; Appendix A). O*NET provides task importance; occupation wages and employment come from OES (2019 vintage; relative occupational wages are highly persistent, and scripted access to the current vintage is blocked —

²This resolves an apparent tension between two bounds that coincidentally round to the same number: the Path-A verification lower bound at our baseline incidence is 0.58%, and the Path-B bound using Brynjolfsson et al. (2025)’s 15% ATE is 0.62% at the same incidence. They are different objects — in particular, Path B carries no ω factor because the experiments are run in work settings, whereas the Path-A bound has already been multiplied by the market-work share — and the two nonetheless land within four basis points of each other; the proximity is informative, not circular.

a logged deviation from our locked specification). The estimation sample is 2,701 task-occupation pairs: 2,063 distinct tasks across 457 occupations, covering 74.1% of the 2019 wage bill. Wage-bill weights apportion each occupation’s wage bill to its tasks by importance; tasks outside the sample count as zero savings in all aggregates, which is conservative.

Protocol. Following the practice of the broader research program, every analysis specification in this paper was locked in a project log before estimation, with deviations disclosed (the OES vintage above; one corrected measurement object documented in Appendix A). We refer to this as specification locking rather than pre-registration, as the log is internal; it is included in the replication package.

3.2 The savings input and its validation

Per-instance savings are $s_j = 1 - t_j^{AI}/t_j^H$, where t^H is the estimated human-only completion time and t^{AI} the with-AI time. The mean is 90.0% and the median 91.2%: the median task would take a human about two and a half hours and takes about fourteen minutes with the model.

These time estimates are model-generated, which demands validation. Four checks. First, units and magnitudes: Anthropic’s own report states speedup factors of 9–12 $\times$ by task education level, which is 89–92% savings — matching our computed distribution. Second, stability: among 1,876 tasks measured in both the November 2025 and February 2026 weeks, per-task savings correlate 0.82, and the aggregate ceiling moves by -0.2% ; the input is a stable object, not weekly noise. Third, internal consistency: our automation/collaboration aggregate (47%/53%) matches the report’s published split (45%/52%) under slightly different exclusions. Fourth, and most demanding: on the three task domains where randomized evidence exists, our measured per-instance savings are 89.7% (writing), 87.1% (coding), and 85.9% (customer support) — essentially uniform — against experimental ATEs of 40% (Noy and Zhang, 2023), 56% (Peng et al., 2023), and 15% (Brynjolfsson et al., 2025). The uniformity is the point: the gap between per-instance savings and ATEs does not track domain difficulty; it is the signature of instance selection, present everywhere and largest where the experiment is most representative of full workloads (customer support, where treated agents handle *all* tickets). This motivates Path B.

3.3 Headline magnitudes

Table 1 summarizes the measured inputs. Median human-to-AI cost ratios (occupational wage $\times$ human time over inference cost) are 636, with an interdecile range of 231 to 1,439. The success rate averages 75.6% (yes/(yes+no) among classified outcomes; including unclassified outcomes in the denominator lowers the waterfall’s upper bound from 15.4% to 11.4% — a robustness we report, not a choice we lean on). The wage-bill-weighted work share is 50.3% (39.4% usage-weighted over distinct tasks): personal use and coursework account for roughly half of how this technology is

Table 1: Measured inputs (November 2025 baseline week)

Input	Value	Acemoglu (2024) analog
Per-instance time savings (mean)	90.0%	27% (imputed from two experiments)
Task success rate	75.6%	—
Market-work share of usage (wage-bill wtd.)	50.3%	—
Delegation share of usage (wage-bill wtd.)	42.6%	—
Wage-bill coverage of measured tasks	74.1%	19.9% exposure $\times$ 23% viable
Median human/AI cost ratio	636 $\times$	—

Notes: 2,701 task-occupation pairs (2,063 tasks, 457 occupations). Savings = $1 - t^{AI}/t^H$ from AEI duration estimates, clipped at zero. Success excludes unclassified outcomes (robustness in text). Work and delegation shares are wage-bill-weighted means of task-level shares — the same objects that enter the waterfall.

actually used.³ The wage-bill-weighted delegation share is 42.6%.

4 Cost No Longer Binds

The historical regularity is that adoption cost governs diffusion speed: across 30 technologies over four centuries, the elasticity of diffusion speed with respect to the cost barrier at launch is approximately -1 (Comin and Hobijn, 2010; Anonymous, 2026). If that regularity applied to AI at the task level, tasks with larger cost-parity gaps should see disproportionate usage, and falling prices should visibly reallocate usage. Neither happens, and the design is strong enough for the null to be informative.

4.1 Cross-section: the gradient is capability in disguise

Within occupations, regressing log relative usage (usage normalized by the task’s economic footprint — importance times employment share — without which the regression mechanically recovers the wage-bill distribution) on the log parity gap gives an elasticity of $+0.215$ (s.e. 0.063): tasks where AI’s cost advantage is larger are indeed used more (Table 2, Figure 1). Adding a single capability control — the task success rate — renders it statistically indistinguishable from zero: -0.030 (s.e. 0.068). Cost advantage and capability are collinear across tasks; the cross-section cannot apportion the gradient between them, and we draw no conclusion about cost’s role from it.

4.2 Panel: a 92% price collapse moves nothing

The panel resolves the collinearity. Across the five vintages, tasks were assigned time-invariant price tiers by task size; commodity-tier output prices fell 92% over the window while frontier prices were flat, so tasks face differential price paths. In first differences — where any fixed task attribute,

³Chatterji et al. (2025) report the non-work share of ChatGPT messages has grown beyond 70%. Claude’s professional skew makes our 50% plausible as an upper bound on the market-work share of consumer AI usage generally.

Table 2: Task-level usage and the cost-parity gap (within-occupation)

Specification	β	s.e.	95% CI	N
No controls	+0.215	0.063	[+0.092, +0.338]	2,609
Task-success control (primary)	−0.030	0.068	[−0.163, +0.104]	2,211
Unnormalized usage (diagnostic)	+0.187	0.063	[+0.063, +0.310]	2,609

Notes: OLS with occupation fixed effects (absorbed by demeaning; singleton occupations dropped), standard errors clustered by occupation. The parity gap is $\log(\text{human cost} / \text{AI cost})$ per task; the AI-cost term is proportional to with-AI completion time, so a uniform rescaling of token prices shifts the intercept, not β .

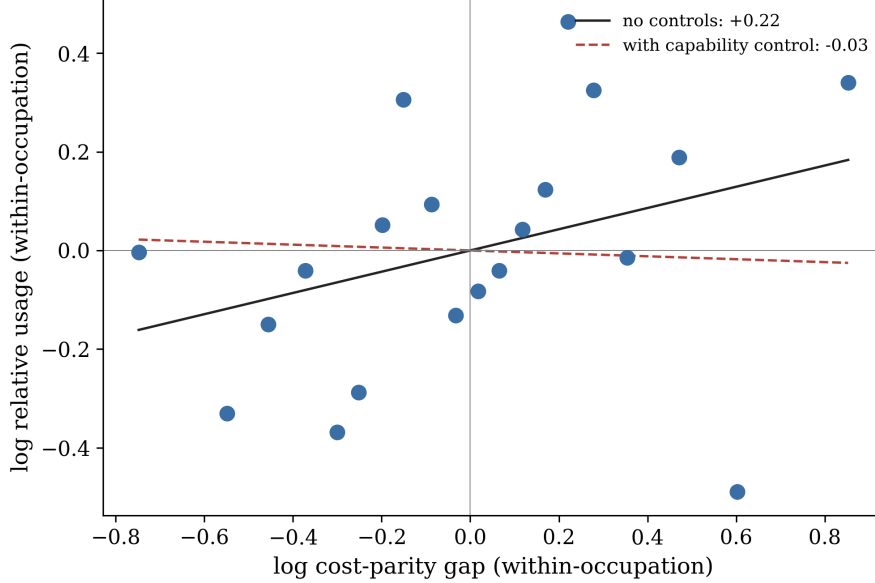


Figure 1: Within-occupation binned scatter of task usage against the cost-parity gap, with and without the capability control.

including capability, nets out — the price elasticity of usage composition is -0.009 , with 95% confidence interval $[-0.030, +0.012]$ (1,869 balanced tasks, 7,476 task-pair observations). Because the price shock varies at the tier level, task-clustered standard errors are anti-conservative (Moulton, 1990); our primary inference is a tier-assignment permutation test, $p = 0.445$. The design’s minimum detectable elasticity at conventional power is 0.031: **the historical benchmark of -1 is more than thirty times the detectable effect, and nothing is there.** Leave-one-vintage-out estimates range from -0.007 to -0.038 (Appendix Table 3); a sparse-pricing composition artifact affecting the early vintages is documented in Appendix A.

The reading is not that prices never mattered; it is that at parity ratios of several hundred, price is far past the adoption margin. AI is the first technology in the diffusion literature’s range to launch already below its cost barrier, and the cost-diffusion law goes quiet exactly where the framework says it should. What remains to govern diffusion is capability and the institutions of deployment — the subjects of the rest of the paper.

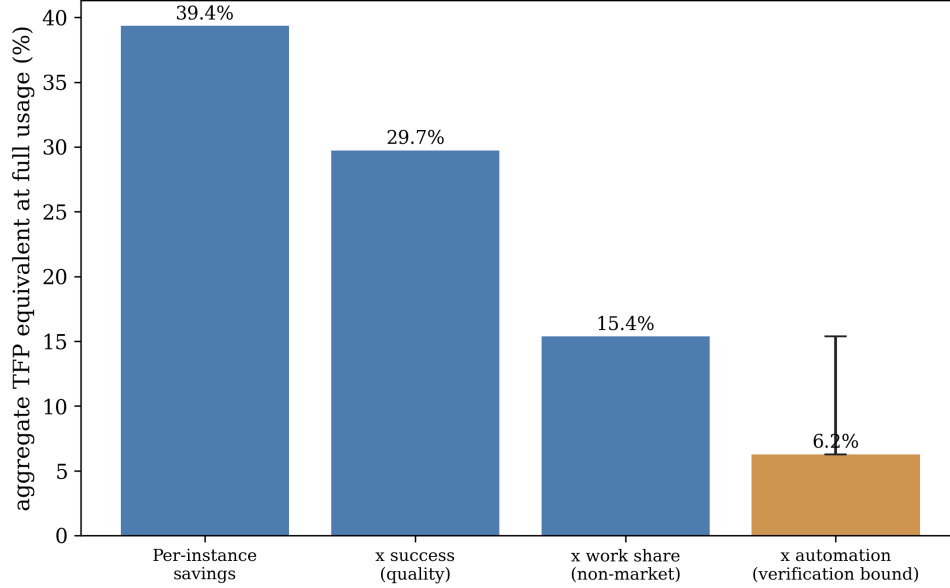


Figure 2: The measured waterfall at full usage ($\iota = 1$). Each stage multiplies the previous by a measured task-level wedge and aggregates through wage-bill weights. The error bar is the verification band.

5 From 90% Savings to 1% TFP: The Waterfall

5.1 Path A: measured wedges

Figure 2 is the paper’s central exhibit. Start from measured per-instance savings aggregated through wage-bill weights at full usage: a ceiling of 39.4% of the covered wage bill in TFP-equivalent terms — an absurd number, and deliberately so, since it embeds maximal selection and no wedges; its only role is to anchor the decomposition. Multiplying by task success rates lowers it to 29.7%. Multiplying by the market-work share lowers it to 15.4% — the single largest measured wedge after adoption itself: roughly half of usage produces surplus that national accounts will never see.⁴ The verification band spans 6.2% to 15.4%: at the lower bound only delegation-mode usage realizes savings; at the upper bound collaborative-mode usage realizes them fully.

Incidence then scales the band. Rather than committing to a single adoption number, Figure 3 (right panel) reports realized TFP as a function of the AI work-hours incidence ι . Bick et al. (2024) measure 1–5% of U.S. work hours as AI-assisted; a coarser construction from firm adoption (Bonney et al., 2024) times within-firm worker use gives 9.3%. At the survey range the realized band is 0.06–0.8%; at the firm-based construction it is **0.6% to 1.4%**, which we quote as the headline.

⁴This has an aggregate-measurement corollary: a technology whose usage is half non-market can transform daily life while leaving a correspondingly attenuated TFP footprint.

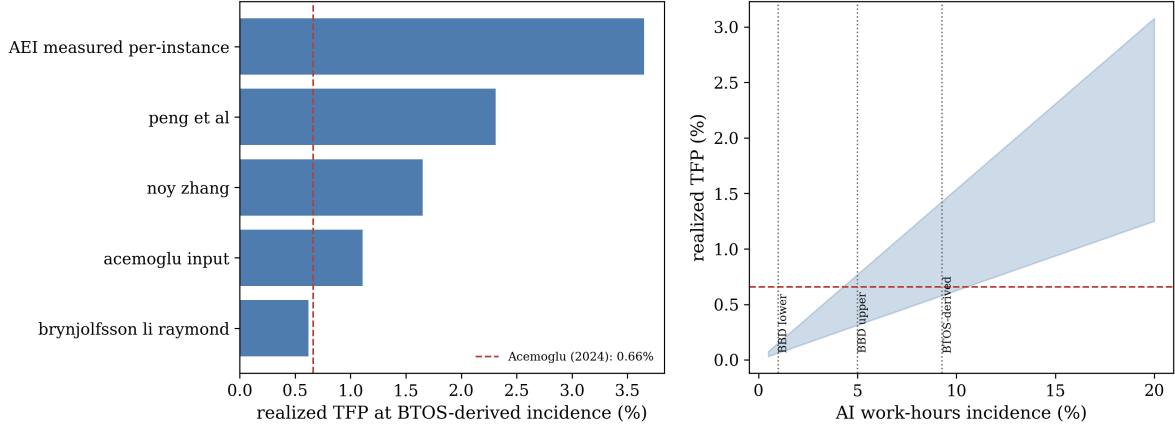


Figure 3: Left: realized TFP at baseline incidence under alternative savings inputs (Path B grid and the measured Path A input). Right: the Path-A band as a function of work-hours incidence, with measured candidates marked.

5.2 Path B: experiment-anchored bounds

Path B replaces measured per-instance savings (and with them the quality and verification wedges, which experiments internalize) with experimental ATEs (Figure 3, left panel). The grid spans Brynjolfsson et al. (2025)’s 15%, Acemoglu (2025)’s 27% calibration — which is exactly the average of Noy and Zhang and Brynjolfsson et al.’s original estimates — Noy and Zhang (2023)’s 40%, and Peng et al. (2023)’s 56%. At the baseline incidence these give 0.62%, 1.11%, 1.65%, and 2.31%. The conservative corner — the most conservative field experiment — lands on Acemoglu (2025)’s 0.66% almost exactly. His bound is not a competing estimate; it is what the measured calculation returns when its most optimistic input is replaced by its most conservative credible value.

5.3 Reconciliation with independent measurements

Three independent methodologies now measure pieces of this object, and they cohere once the incidence margin is explicit. Our per-instance savings (90%) times Bick et al.’s assisted-hours incidence (1–5%) implies time savings of 0.9–4.5% of work hours; Bick et al. (2024) directly measure 1.4%, and Humlum and Vestergaard (2025) measure 3% among Danish users in exposed occupations — both inside the implied range. At the labor share, Bick et al.’s direct figure corresponds to a TFP contribution of roughly 0.8% — inside our band. Humlum and Vestergaard (2025)’s null on earnings is likewise consistent: gains of this size, divided between many margins, are not detectable in individual labor-market outcomes.

6 The Deployment Headroom

If price does not govern diffusion and capability is improving on its own schedule, the policy-relevant margin is deployment: the infrastructure that turns interactive assistance into delegated

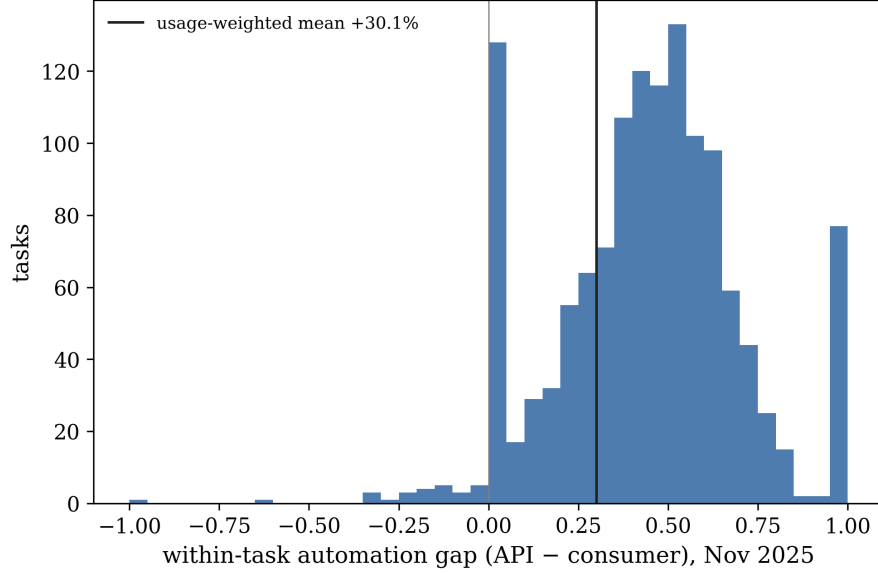


Figure 4: Distribution of the within-task automation gap (API – consumer), November 2025.

production. The AEI’s parallel coverage of consumer and API traffic measures this margin directly.

Within the same O*NET task in the same November 2025 week, API usage is dramatically more delegated: the usage-weighted within-task gap in automation shares is **+30.1 percentage points** (median +45.5; 89% of the 1,322 common tasks are more automated on the API), and the aggregate automation shares are 84.9% versus 47.2%. We emphasize what this is and is not. It is not a treatment effect of scaffolding: firms select which tasks and instances to run through production deployments, and identical task text does not mean identical instances, users, or stakes. It is a measured *headroom* — an upper bound on what deployment infrastructure currently achieves relative to interactive use, holding the task taxonomy fixed. Mechanically, moving work usage from the consumer pattern to the API pattern would raise the waterfall’s verification lower bound from 0.6% to roughly 1.0% with no change in model capability.

Two decompositions discipline the interpretation (Appendix Table 4). First, a shift-share across vintages: aggregate automation shares *declined* on both platforms between August 2025 and February 2026 (−5.7 points consumer, −6.1 API), and on both platforms the decline is within-task (−3.8, −5.5) rather than compositional (−1.9, −0.6). The verification tax is not falling on its own. Second, a changes-on-changes regression of within-task automation movements: on the consumer platform, tasks whose instances grew longer became less automated (coefficient −0.047, $p = .035$) — instance-complexity creep — while task-level user-skill shifts do nothing (+0.004, $p = .59$); on the API the complexity coefficient *flips sign* (+0.056, $p = .03$): where scaffolding absorbs them, bigger jobs become more delegated, not less. That sign flip is the mechanism evidence: deployment infrastructure converts exactly the workload growth that pushes interactive users into supervision.

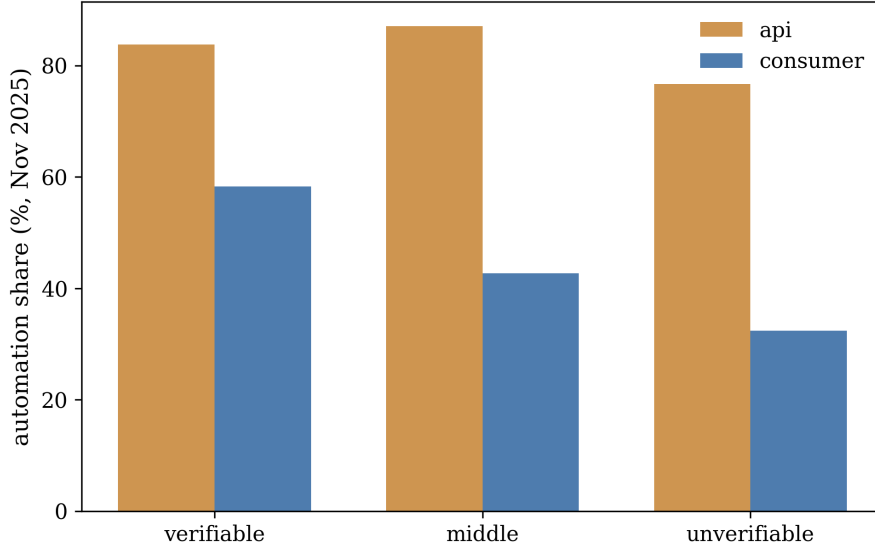


Figure 5: Automation share by machine-checkability of task outputs, November 2025.

7 The Verifiability Gradient

Why does delegation stop where it stops? The natural theory is verification: humans stay in the loop where outputs cannot be checked mechanically. On the consumer platform, delegation is monotone in machine-checkability — the pattern verification theories predict: classifying tasks by machine-checkability of outputs (a transparent keyword rubric; order-flip robustness in Appendix B), automation shares are 58.3% for verifiable tasks, 42.7% in the middle, and 32.4% for unverifiable tasks (Figure 5).

The API breaks the theory’s sharpest prediction. If verification technology drove delegation, the API-consumer headroom should concentrate in verifiable domains. It does the opposite: the headroom is *largest for unverifiable tasks* (+44.3 points versus +25.6 for verifiable), and the API gradient is nearly flat and non-monotonic (83.8% verifiable, 87.1% middle, 76.7% unverifiable). Deployment context substitutes for machine-verifiability: production settings delegate broadly — moving verification downstream, into sampling, process controls, or risk tolerance — rather than expanding delegation only where outputs are checkable in-loop.

The wage-bill geography makes this first-order. Machine-verifiable tasks are 14.7% of the covered wage bill; unverifiable tasks 11.5%; the ambiguous middle 73.8%. A verification-led diffusion would saturate a seventh of the wage bill and stall. The measured pattern — deployment buying delegation wholesale across the middle mass — is the one consistent with the headroom reaching most of the economy.

8 External Validity and Limitations

Selection is the design, not a footnote. Every usage-based quantity conditions on adoption of one provider’s models and on the instances users chose to bring. Our response is architectural: the ceiling-to-realized gap is reported as the measurement of selection; experimental ATEs provide the selection-free anchor (Path B); and the savings-validation table shows the selection wedge is uniform across the domains where experiments exist, rather than an artifact of domain mix.

One provider. Claude’s user base skews professional. [Chatterji et al. \(2025\)](#)’s ChatGPT estimates suggest our 50% market-work share is, if anything, an upper bound for consumer AI usage generally — which would push the non-market wedge further and our band down, not up.

Model-estimated durations. The savings input is generated by classifiers, not stopwatches. Section 3.2 stacks four validations; the residual risk is a level bias common to both duration estimates, which the ratio partially nets out, and which Path B does not use at all.

Single-week vintages. Each vintage is one week of traffic; no within-vintage replicate exists to estimate weekly sampling noise. Cross-vintage stability (savings correlation 0.82; wedge movements of a point or two over months) bounds the concern but cannot eliminate it.

Static accounting. The framework prices task-level cost savings; it omits general-equilibrium reallocation, new-task creation (we measure the taxonomy’s non-coverage of usage at a stable $\sim 7\%$, suggesting modest new-task mass so far), deepening of AI capital, and any deflator mismeasurement of AI-intensive output. These are directions, not corrections we can sign.

9 Conclusion

The inputs of AI’s aggregate productivity arithmetic are now measurable, and measuring them reorders the policy conversation. Per-instance savings are an order of magnitude larger than the experimental calibrations in use — and aggregate implications are nevertheless about what those calibrations implied, because the wedges between an assisted instance and a unit of national output are large, measurable, and mostly ignored: half of usage is not market work; a quarter of attempts fail; half of work usage keeps a human in the loop; and only a few percent of work hours are assisted at all. The binding constraints, in order of measured leverage: adoption incidence (the entire ceiling-to-realized gap), deployment infrastructure (thirty points of within-task headroom, operating broadly and precisely where outputs resist mechanical checking), verification practice (static on its own), then model capability, whose marginal leverage we do not measure and treat as the residual — and price nowhere, having already fallen below every margin that mattered.

The anatomy is static, but its inputs need not be: each wedge has a measured speed, and a companion projection exercise in the replication materials propagates them forward. Three of

its outputs are worth recording here because they are testable on the same data that will update this paper. First, calibrating today’s realized band against the task-reliability frontier implies that adopting firms already use AI on nearly all of the task-hours where it is currently reliable; growth must come from the extensive adoption margin and from verification, not from deeper use of feasible tasks. Second, the projected acceleration paths cluster at two dates — 2027 if capability-horizon growth converts to realized output at its measured doubling rate, 2030 if instead the verification and quality wedges begin moving — and a quiet 2027 in utilization-adjusted TFP rejects the first channel within about eight quarters. The first vintage update after drafting already moved one of the two: wage-bill-weighted task success rose from 76.0 to 79.7 percent between the November 2025 and February 2026 releases, several times faster than the projection’s capability-unlock parameterization, while the automation share stayed inside its noise band — the quality wedge is moving before the verification wedge. Third, under the conservative corner (flat wedges, slow diffusion) the realized contribution plateaus below 1.5 percent and never produces a detectable growth increment: the transition stays permanently inside the measurement wedges documented above. Two companion extensions discipline the projections from opposite sides: letting per-instance savings decay toward the experimental estimates as adoption broadens roughly halves the projected long-run levels without touching the 2027 test, and recomputing the AI capital contribution at an income-share-consistent elasticity (capital expenditure of 1.3 percent of GDP supports a contribution near 0.2 points, not the 1.75 points a conventional 0.05 elasticity implies) shrinks the investment-phase drag on measured TFP by nearly an order of magnitude — which cuts both ways, since a shallow J-curve trough also promises a smaller mechanical rebound.

For productivity measurement, one descriptive note. U.S. utilization-adjusted TFP growth in 2023–2026 ran about 0.4 points above its 2010–2019 average (Fernald, 2014). Our realized-now band implies a contribution of 0.06–0.14 points per year if spread over a decade. The acceleration, if it persists and if any of it is AI, is at or above what the measured micro-anatomy currently explains — a gap for which the deployment headroom, not further model capability, is the natural candidate.

References

- Acemoglu, Daron**, “The Simple Macroeconomics of AI,” *Economic Policy*, 2025, 40 (121), 13–58. NBER Working Paper 32487.
- Anonymous**, “From Cost Decline to Productivity Transition: How Falling Barriers Accelerate Adoption and Destabilize Growth Regimes,” Technical Report, Working Paper 2026.
- Anthropic**, “The Anthropic Economic Index Report: Economic Primitives,” <https://www.anthropic.com/research/anthropic-economic-index-january-2026-report> 2026. Data releases at <https://huggingface.co/datasets/Anthropic/EconomicIndex>.

- Bick, Alexander, Adam Blandin, and David J. Deming**, “The Rapid Adoption of Generative AI,” Working Paper 32966, National Bureau of Economic Research 2024.
- Bonney, Kathryn, Cory Breaux, Cathy Buffington, Emin Dinlersoz, Lucia Foster, Nathan Goldschlag, John Haltiwanger, Zachary Kroff, and Keith Savage**, “Tracking Firm Use of AI in Real Time: A Snapshot from the Business Trends and Outlook Survey,” Working Paper CES-WP-24-16, Census Bureau, Center for Economic Studies 2024.
- Brynjolfsson, Erik, Danielle Li, and Lindsey Raymond**, “Generative AI at Work,” *Quarterly Journal of Economics*, 2025, 140 (2), 889–942.
- Chatterji, Aaron, Thomas Cunningham, David J. Deming, Zoë Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman**, “How People Use ChatGPT,” Working Paper 34255, National Bureau of Economic Research 2025.
- Comin, Diego and Bart Hobijn**, “An Exploration of Technology Diffusion,” *American Economic Review*, 2010, 100 (5), 2031–2059.
- Dell’Acqua, Fabrizio, Edward McFowland III, Ethan Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Kraymer, François Candelon, and Karim Lakhani**, “Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality,” *Harvard Business School Working Paper 24-013*, 2023.
- Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock**, “GPTs are GPTs: Labor Market Impact Potential of LLMs,” *Science*, 2024, 384 (6702), 1306–1308.
- Fernald, John G.**, “A Quarterly, Utilization-Adjusted Series on Total Factor Productivity,” *FRBSF Working Paper 2012-19*, 2014.
- Handa, Kunal, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus et al.**, “Which Economic Tasks are Performed with AI? Evidence from a Million Anthropic Claude Conversations,” Technical Report, Anthropic 2025. Anthropic Economic Index.
- Hulten, Charles R.**, “Growth Accounting with Intermediate Inputs,” *Review of Economic Studies*, 1978, 45 (3), 511–518.
- Humlum, Anders and Emilie Vestergaard**, “Large Language Models, Small Labor Market Effects,” Working Paper 33777, National Bureau of Economic Research 2025.
- Moulton, Brent R.**, “An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units,” *Review of Economics and Statistics*, 1990, 72 (2), 334–338.
- Noy, Shakked and Whitney Zhang**, “Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence,” *Science*, 2023, 381 (6654), 187–192.

Peng, Sida, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer, “The Impact of AI on Developer Productivity: Evidence from GitHub Copilot,” Technical Report 2302.06590, arXiv 2023.

Svanberg, Maja, Wensu Li, Martin Fleming, Brian Goehring, and Neil Thompson, “Beyond AI Exposure: Which Tasks are Cost-Effective to Automate with Computer Vision?,” *MIT FutureTech Working Paper*, 2024.

A Data Construction and Measurement Details

Vintages and schema. Consumer (Claude.ai) and first-party API raw files for the weeks of 2025-08-04/11, 2025-11-13/20, and 2026-02-05/12, plus task-share tables for the two early windows, all from Anthropic’s public releases. The February 2026 release appends a `::value` suffix to time-facet cluster names; our extraction strips it. Success outcomes carry `::yes/::no/::not_classified` suffixes; the success rate is `yes/(yes+no)`, with the unclassified-inclusive variant reported in the text.

Units. The AEI documentation does not state units for the duration facets. Three observations pin them: human-only times (median 2.6, maximum 14) are implausible as minutes and natural as hours; with-AI times (median 14.4) match conversation lengths in minutes; and the implied savings distribution reproduces the report’s published 9–12 $\times$ speedups. We convert human-only times to minutes at extraction and document the inference.

O*NET merge and version drift. 13.7% of AEI task names fail exact-text match to O*NET 30.1 statements. A fuzzy-match diagnosis on a sample of unmatched names finds median best-match similarity 0.96, with 55% above 0.9: version drift between Anthropic’s O*NET vintage and ours, not new tasks. The honest measure of work outside the taxonomy is the AEI’s own unclassified share of task usage, stable at 6.5–8.1% across vintages.

Within-SOC duplicates. A six-digit SOC containing several eight-digit O*NET codes can list one task statement repeatedly; such rows are exact duplicates within the fixed-effect group and are dropped (six rows). A task legitimately belonging to several occupations is retained once per occupation; wage-bill weights are normalized within occupation, so this does not inflate aggregates.

Quoted aggregate shares. All aggregate wedge shares quoted in the paper are wage-bill-weighted means of task-level shares over the estimation sample — by construction the same objects that enter the waterfall — with usage-weighted distinct-task variants in the replication output. An earlier internal memo quoted shares computed from raw facet totals (e.g., a 46.3% work share); these mixed samples and are superseded.

Tier price paths. Tasks are assigned commodity/mid/frontier tiers by terciles of human-only time; tier prices are the latest output-token price at or before each vintage date from a 15-row historical pricing table. The table’s sparsity creates one composition artifact (the representative mid-tier model changes between the first two vintages); leave-one-vintage-out estimates bound its influence (Table 3).

Table 3: Panel first differences: leave-one-vintage-out

Dropped vintage	β	s.e.
v1 (Dec 24–Jan 25)	−0.038	0.019
v2 (Feb–Mar 25)	−0.034	0.013
Aug 25	−0.012	0.011
Nov 25	−0.019	0.011
Feb 26	−0.007	0.012

Notes: cluster (task) standard errors shown; tier-level treatment makes these anti-conservative, and the permutation test remains the primary inference throughout.

Table 4: Why within-task automation moved (Nov 2025 → Feb 2026)

Regressor	Consumer		API	
	coef.	p	coef.	p
Δ log instance size	−0.047	.035	+0.056	.034
Δ user education	+0.004	.592	−0.034	.000
Δ log usage	+0.061	.000	−0.066	.006

Notes: WLS in task-level changes, weighted by mean usage, HC1 standard errors. Outcome: change in the task’s automation share.

B Additional Results

Verifiability rubric. Tasks matching both keyword lists ($\sim 2\%$ of classified tasks) are assigned by check order; flipping the order moves the consumer verifiable-class share by 1.6 points and every other cell by under a point — no qualitative statement changes against a 26-point gradient. The rubric is heuristic; the consumer-level gradient (26-point spread) is far larger than any classification ambiguity.

Trajectories. Aggregate delegation shares by vintage (42.6, 43.1, 51.6, 47.2, 45.9% on the consumer platform; the first two points use the earlier releases’ aggregate interaction-type tables and are not strictly method-consistent with the rest) and the taxonomy non-coverage series (8.1, 6.5, 7.0%) are in the replication output.